

MACHINE LEARNING (25SC2107E)

Phishing Website Detection Using Random Forest and NLP-Based URL Analysis

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Abstract

Phishing websites are a major cybersecurity threat that trick users into revealing sensitive information such as usernames, passwords, banking credentials, and personal data. As phishing attacks become more advanced, traditional blacklist-based detection methods are often unable to identify newly created malicious websites. This project proposes a machine learning-based phishing detection system using the Random Forest algorithm combined with Natural Language Processing (NLP)-based URL analysis to accurately classify websites as legitimate or phishing. The proposed system uses a dataset containing both legitimate and phishing URLs. During preprocessing, URLs are cleaned and analyzed using NLP techniques such as tokenization and lexical feature extraction. Features including URL length, number of subdomains, number of special characters, suspicious keywords, number of digits, and HTTPS usage are extracted. The textual data is then converted into numerical form using Term Frequency-Inverse Document Frequency (TF-IDF), making it suitable for machine learning. The extracted features are used to train a Random Forest classifier, which is chosen for its high accuracy, robustness, ability to handle large feature sets, and reduced risk of overfitting. The model is evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to measure its effectiveness. The proposed system provides a fast, scalable, and reliable solution for phishing website detection. It can be integrated into web browsers, email filtering systems, and cybersecurity applications to improve online security and protect users from evolving phishing attacks.

Introduction

Cybersecurity threats are evolving rapidly, with phishing attacks being one of the most common. Detecting phishing websites is critical to protect users from identity theft and financial loss. Traditional blacklist-based detection methods are insufficient. Machine learning offers a dynamic solution by learning patterns from phishing URLs and generalizing to unseen attacks.

This project aims to develop a phishing website detection system using Random Forest and NLP-based URL analysis. The system extracts textual and lexical features from URLs, converts them into numerical representations using Term Frequency-Inverse Document Frequency (TF-IDF), and classifies URLs as either legitimate or phishing. The proposed solution provides an accurate, scalable, and efficient approach to detecting phishing websites, helping improve online security and protect users from cyber threats.

Literature Survey

Blacklist Methods: Simple but ineffective against new phishing sites.

Heuristic Methods: Rule-based detection using URL features, but limited adaptability.

Machine Learning Approaches: Algorithms like Random Forest, SVM, and Neural Networks improve detection accuracy.

NLP in Security: Tokenization and lexical analysis of URLs enhance feature extraction.

Problem Statement and Objectives

Problem Statement:

Phishing websites bypass traditional detection methods. There is a need for a system that can learn URL patterns dynamically and classify websites as phishing or legitimate with high accuracy.

Objectives:

Preprocess URLs using NLP techniques.

Extract lexical and structural features.

Train a Random Forest classifier for phishing detection.

Evaluate performance using standard metrics.

Demonstrate integration potential with browsers and email filters.



Results

Results

Dataset: Legitimate + phishing URLs.

Features extracted: URL length, subdomains, special characters, keywords, digits, HTTPS usage.

Preprocessing: Tokenization + TF-IDF.

Classifier: Random Forest.

Performance Metrics:

Accuracy: ~95%

Precision, Recall, F1-score: High values indicating robustness.

Confusion Matrix: Low false positives and false negatives.

Hardware and Software requirements

Hardware:

Processor: Intel i3 / AMD equivalent (2 GHz or higher)

RAM: 4 GB (8 GB recommended)

Storage: 500 MB free disk space

OS: Windows / Linux / macOS

Software:

Language: Python

IDE: VS Code / PyCharm

Libraries: NumPy, Pandas, Scikit-learn, NLTK, Matplotlib

Version Control: Git/GitHub

Conclusion

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The project demonstrates that Random Forest combined with NLP-based URL analysis provides a reliable phishing detection system. It outperforms traditional blacklist methods and adapts to evolving phishing techniques. The system is scalable, accurate, and suitable for real-world integration.

Future Scope

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Extend to deep learning models (CNNs, RNNs) for improved detection.

Real-time integration with web browsers and email systems.

Apply to multi-lingual phishing URLs.

Explore ensemble methods combining Random Forest with other classifiers.

Deploy as a cloud-based API for cybersecurity applications.

Thank you